

The Future of Networking, and the Past of Protocols - Scott Shenker

Scott Shenker argues that the future of networking lies in finding the right abstractions and discusses the value proposition of Software Defined Networks (SDNs). Scott discusses the "why", i.e. the philosophy of SDNs, rather than the "what". The speaker starts off by pointing out the importance of the layered architecture of the internet, which enables innovation at each layer, but calls it an academic failure as it is an artifact rather than a discipline.

The speaker argues that the other fields of Computer Science, like operating systems and databases, have some basic principles, are easy to manage and continue to evolve. Networks, on the other hand is just a big bag of protocols, which is difficult to manage, and evolve very slowly. New control requirements have led to a lot of complexity in networks. Shenker states that the ability to master complexity is not the same as the ability to extract simplicity; we are still mastering complexity and never made the transition to extracting simplicity.

The speaker presents the example of programming and argues that abstractions are the key to extracting simplicity. Abstractions lead to interfaces which makes the programs modular, which is why we are able to get it to work and maintain it. In networking, layers provide abstractions, but we do not have any abstractions in the control plane. We need abstractions for **1.** the distributed state to deal with lack of communication guarantees, **2.** simplifying configuration of physical devices and **3.** a general forwarding model to operate within the given network level protocol.

A network graph API can provide abstraction over the global network view, which would allow programmers to use graph algorithms like Dijkstra. For specification abstraction, the control program should express the desired behavior through network virtualization, rather than being responsible for implementing that behavior on the network. For the flexible forwarding model, the abstraction should not constrain the control program but it should hide the details of the underlying hardware.

SDNs provide these three abstractions, and their value proposition is the same as that of the Internet's core architecture, i.e. innovation. SDN provides clean separation of concerns and comprises of a control program, which configures the abstract network view, a network virtualization layer which configures the global network view based on the abstract network view, which is then communicated to the physical devices by the Network OS. Shenker argues that the key value proposition of SDNs is innovation.

The talk was very informative, especially to newcomers like myself. One interesting aspect of the talk is its parallels to modern day cloud computing, particularly with the idea of abstraction. AWS for instance, supports cloud formation templates, where the user can programmatically define the various components of the infrastructure and their access policies. The template can then be uploaded to the AWS console and AWS will take care of the resource provisioning and applying actual configuration. With SDNs, such high level, programmatic control of networks, without worrying about the hardware implementations becomes possible.